João Ferreira

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Profile

MSc student in Electrical and Computer Engineering with hands-on experience in embedded systems, IoT, and smart building systems. Skilled in end-to-end IoT platform development, ESP32 sensor firmware to MQTT pipelines, real-time dashboards, and home automation, with additional experience in electric vehicle communications and TinyML for resource-constrained environments.

Education

MSc in Electrical and Computer Engineering 09/2024 – Present
Universidade de Trás-os-Montes e Alto Douro (UTAD) *Expected completion: July 2026*

BSc in Electrical and Computer Engineering 10/2021 – 07/2024
Universidade de Trás-os-Montes e Alto Douro (UTAD) *Vila Real, Portugal*

Cooperative International Student Project 09/2023 – 02/2024
FH Technikum Wien *Vienna, Austria*

- Collaborated in a multicultural team with universities from Finland, Portugal, and Austria, strengthening scientific documentation and international communication skills.
- Conducted performance data analysis to optimize rotor designs and electrical systems.
- Developed practical skills in 3D printing, wind tunnel testing, and the integration of generators, inverters, and batteries into renewable energy systems.

Professional Experience

Intelligent Systems Engineer – Casas em Movimento 4.0 Project 09/2026 – Present
CeM – Casas em Movimento *Vila Real, Portugal*

- Joined CeM to strengthen its technical structure in the field of intelligent systems, supporting the execution and development of the Casas em Movimento 4.0 project.
- Contribute to the planning and development of digital twin models for real homes.
- Develop an application to support these models and the interconnectivity between the home, the user, and mobile devices.

Researcher – A-MoVeR Project 10/2025 – 06/2026
UTAD / AJP Motorcycles / Continental Advanced Antenna *Vila Real, Portugal*

- Conducted research within the A-MoVeR project for electric motorcycle development and validation.
- Designed a Hardware-in-the-Loop validation infrastructure and an ESP32-based onboard telemetry node for the Fulgora electric motorcycle.
- Developed CAN bus datalogging and BLE-based wireless transmission solutions.
- Evaluated TinyML models for predictive range estimation in embedded environments.

Projects

MonitorAR – Smart Room Monitoring & Control Platform 03/2025 – Present
Personal Project *Remote*

- Designed and deployed a full-stack IoT platform for intelligent room monitoring: ESP32 sensor nodes (BME680 for air quality; LD2410 mmWave radar for presence detection), MQTT broker, InfluxDB/Grafana data pipeline, and a custom 3D web application as the main cockpit.
- Built a Room State Engine in Python for automated presence detection and contextual space adaptation, integrated with Home Assistant for smart device control.
- Developed a physical control interface using a Raspberry Pi Zero 2 W with an e-ink display for local interaction.
- Deployed the full stack on self-hosted Raspberry Pi infrastructure using Docker and Nginx Proxy Manager.

ESP32 PID Climate Controller 2025
Personal Project *Remote*

- Designed a PID-based indoor humidity control system on an ESP32, with a simulation phase for gain tuning validated against a mathematical room model.
- Implemented real-world control of a Gree Inverter AC unit via Wi-Fi using a multi-level control strategy to optimise comfort and energy efficiency.
- Published all simulation data to an MQTT broker (HiveMQ) for real-time monitoring in Grafana.

Skills

Programming Languages: C, C++, C#, MATLAB/Simulink, Python
Operating Systems & Software: Linux (Ubuntu/Debian), Docker, Portainer
Hardware & Embedded Systems: Raspberry Pi, ESP32, I2C, UART
Networking & IoT: MQTT, Mosquitto, InfluxDB, Nginx, Home Assistant, Matter
Machine Learning & Edge AI: TinyML, anomaly detection, intrusion detection systems
Protocols & Networks: CAN Bus, MQTT, UART, I2C, wireless communications

Languages

- **Portuguese**: Native
- **English**: Centro Anglo-Americano (CAA) Advanced - C1; Cambridge English First (FCE) - B2
- **Spanish**: Intermediate

Certificates

- MATLAB Onramp
- MATLAB Fundamentals
- MATLAB Programming Techniques
- MATLAB for Data Processing and Visualization